

# Kaitlyn Forks

## Full Stack Software Engineer in St. Louis, MO

A published peer-reviewed **Biologist turned Software Engineer**, Kaitlyn learned R and Python while researching reptiles and amphibians, using code to analyze ecological data and build predictive models. These foundational skills led her to full-stack development, where she now specializes in lightweight architecture, relational database design, and ensuring scalable applications **using SQL, Node.js, Angular and AWS Cloud Services**.

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## EDUCATION

**LaunchCode**, Software Development Bootcamp — January 2020

**University of Missouri St. Louis**, MA, Science Education — May 2019

**Texas A&M University**, BA, Ecology & Evolutionary Biology — May 2017

## WORK EXPERIENCE

### Software Engineer — Appriss Insights / Equifax

Oct 2022 - December 2024

- **Led the Node.js / Angular front-end development for MOVAN, an internal ETL (Extract, Transform, Load) application** that automates real-time notifications to keep victims informed about changes in perpetrator status.
- **Collaborated with backend engineers to optimize data ingestion and transformation workflows in Python**, improving processing time and system efficiency.
- **Developed front-end components that optimized large-scale data ingestion and transformation**, allowing users to manage high-volume data jobs intuitively.
- **Enhanced data visualization to improve clarity for internal users**, reducing errors in data transformation.
- **Automated form validation and user input verification**, increasing data integrity and reducing manual errors.

### Full Stack Software Engineer — Anchor Yacht Charters

May 2021 - Oct 2022

- **Architected the Boat Owner Calendar and Charter Booking features in Node.js, Angular and Hubspot**, allowing full customization of schedules, eliminating double bookings, and removing employees from the scheduling workflow.
- **Automated payout systems using Stripe**, streamlining financial workflows, reducing payment processing time, and ensuring secure financial data handling.
- **Designed a microservice for tax and fee calculations**, improving system performance and ensuring compliance with local regulations.
- **Integrated Google Maps API to enhance real-time location services**, increasing bookings and customer satisfaction.
- **Worked directly with CEOs and stakeholders** to deliver features that contributed to positive year-over-year growth in charter bookings.

## SKILLS

### PROGRAMMING LANGUAGES

- Javascript
- Typescript
- Python
- HTML/CSS
- Java
- R

### FRAMEWORKS

- Node.js
- Angular
- React
- NestJS
- Next.js
- Flask
- Sequelize ORM

### DATABASES

- MySQL
- SQLite
- PostgreSQL

### Integrations/APIs/Cloud

- AWS Cloud Services
- RESTful APIs
- Stripe
- Google Maps API
- Hubspot & Twilio

### Workflows, Methodologies, etc.

- Consulting
- Startup Experience
- Database Design
- End to End / E2E
- Test Driven Development
- Agile Workflow

## WORK EXPERIENCE CONTINUED

### Junior Software Engineer — Varsity Tutors LLC / Nerdy NYSE

Dec 2020 - May 2021

- **Spearheaded the automation of manual tasks during my role on the business side**, reducing my daily workload by 4 hours through developer tools automation, which led to my transition to the engineering team.
- **Automated the creation of scalable landing pages in YAML by linking a CSV input process**, enabling business teams to generate large batches of B2B marketing pages, improving campaign turnaround times.
- **Integrated Dato CMS to enable non-technical staff to manage landing pages independently**, freeing up the engineering team to focus on SEO optimization and custom reusable components.
- **Focused on improving lead generation and conversion by building and testing reusable front-end components in Ruby on Rails**, contributing to increased engagement across marketing campaigns.

## ADDITIONAL PROJECTS

### Technical Consultant — National Science Foundation, INCLUDES Network

Freelance 2025

Collaborated with an Associate Professor at Washington University in St. Louis to preserve and optimize data from the decommissioned NSF INCLUDES National Network, which supports diversity, equity, and inclusion (DEI) initiatives in STEM.

- **Ethically scraped and organized web data into usable formats**, preserving critical resources for academic research and outreach efforts.
- **Processed and transformed raw HTML data using Python, Pandas, and Beautiful Soup**, ensuring structured and clean datasets tailored to the client's needs.
- **Built a lightweight back-end using Flask** to enable efficient data parsing and organization.
- Designed React-based filterable tables to display key data interactively, improving usability and navigation for researchers.
- **Delivered clean, searchable CSV files and dynamic single-page summaries**, streamlining the client's workflow and making data more actionable.
- **Demonstrated expertise in consulting, full-stack development and data management**, integrating modern technologies to solve real-world problems.

### Lead Full Stack Engineer and Tech Consultant — Spite House Studio Party Games

Freelance 2022

Led Technical Development for a Kickstarter campaign that raised over a quarter of a million dollars to create and distribute a card game based on principles of improvised comedy.

- **Developed the entire e-commerce platform alongside one other engineer, using NestJS, Angular, and SQL** to support secure payments, order fulfillment, and product browsing.
- **Integrated Stripe for payment processing and HubSpot for automating business email workflows**, streamlining customer communications and marketing efforts.
- **Connected distribution and shipping APIs from ShipStation and "Fun Again" Board Game makers to automate order fulfillment and customer service interactions**, reducing manual processing time and ensuring timely deliveries.
- **Built dynamic product carousels, shopping cart logic, and coupon code functionalities**, enhancing the user experience and driving increased sales conversions.
- **Collaborated closely with the CEO and product team to ensure the e-commerce platform aligned with business goals**, resulting in a successful launch and increased revenue.

## RESEARCH & ACADEMIC EXPERIENCE

### Research Assistant — Ecological Systems Laboratory, Texas A&M University

June 2015 - May 2017

- **Developed population models using STELLA and R** to analyze ecological data and predict population trends.
- **Presented research** on trapping methods and fire ecology at national symposia.

### Intern — United States Geological Survey, Gainesville, Florida

July 2016 - August 2016

- **Used R and Python for hierarchical data analysis**, developing code to predict home ranges of tree frogs.
- **Assisted in capture and genetic sampling** of amphibians for research on disease transmission.

## PEER REVIEWED PUBLICATIONS & RESEARCH AWARDS

**Prevalence of *Batrachochytrium dendrobatidis* in Two Sympatric Tree Frog Species, *Hyla cinerea* and *Hyla versicolor*** — *Herpetological Review*, Dec 1, 2016

**A Simple Conservation Tool to Aid Restoration of Amphibians Following High-Severity Wildfires: Use of PVC Pipes by Green Tree Frogs (*Hyla cinerea*) in Central Texas, USA** — *Diversity* 13, no. 12: 649 · Dec 6, 2021

**Applied Biodiversity Conservation Scholar** — National Fish and Wildlife Foundation (NFWF) — 2016

**2nd Place Poster Presentation** — Ecological Integration Symposium — April 2016

**Undergraduate Research Scholar** — Department of Wildlife & Fisheries Sciences, Texas A&M University — May 2016

**Undergraduate Research Grant Recipient** — Texas A&M University — August 2015

## EXTRAS

**Local Comedian:** Regular performer and teacher at The Improv Shop in St. Louis, where I've been on a house team since 2018. I've traveled with my teams to Comedy Festivals locally and nationally, namely as a repeat performer Flyover and Omaha Comedy Festivals.

**Skydiving on My First Flight:** The first time I flew on a plane, I jumped out of it—skydiving to kick off a lifetime of adventure, and 6 months of not knowing “where or when I'd land”.

**Saint Louis Zoo Affiliate:** Spent nights sleeping in the sea lion tunnel and developed a curriculum to help Girl Scouts earn their badges. Worked in the Department of Audience Research and did Participatory Asset Mapping to prepare for the 2027 opening of WildCare Park, a world-class safari park.

**Reptile Girl:** My dad co-wrote the Field Guide to Texas Snakes, and growing up in a house with a snake farm inspired my lifelong love for reptiles and amphibians.

**Former Public School Teacher:** I am a Teach For America Alumni. I served two years as a Middle School Science teacher in the Saint Louis Public School system where I earned my teaching certificate, licensure, and Masters in Secondary Science Education.

**Hobby Fiend:** When I'm not coding, I'm tending to my garden, playing the ukulele, baking bread, working out, painting, drawing, or creating culinary art!